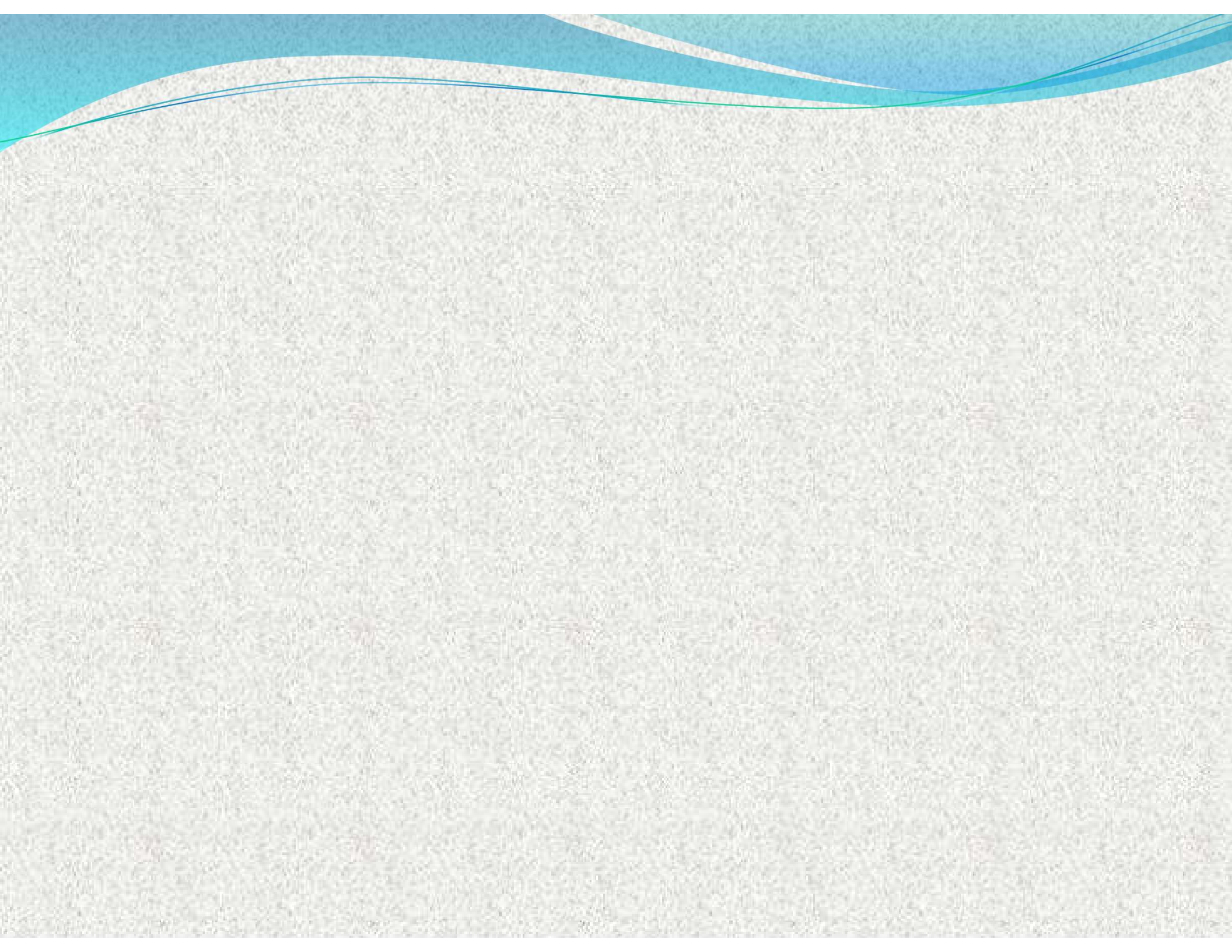




Wave Mechanics

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Advent of the QUANTUM Mechanics.....

The Classical mechanics is not able to explain following-

1. Photo electric effect
2. Black body radiation
3. Compton effect

The most outstanding development in modern science was the conception of the New type of Physics, known as Quantum Mechanics.

Quantum Mechanics is highly successful in explaining the behavior of atoms, molecules and nuclei.

Quantum Mechanics

Max Planck in 1900 proposed that the Energy of black body radiator is distributed in the small packets (quanta) of Energy.

This was the start of the revolution of Physics i.e. the start of Quantum Mechanics.

Quantum Mechanics is a branch of science that deals with discrete, indivisible units of energy called quanta as described by the Quantum Theory.

Main Points of Quantum Theory

There are five main ideas represented in Quantum Theory:

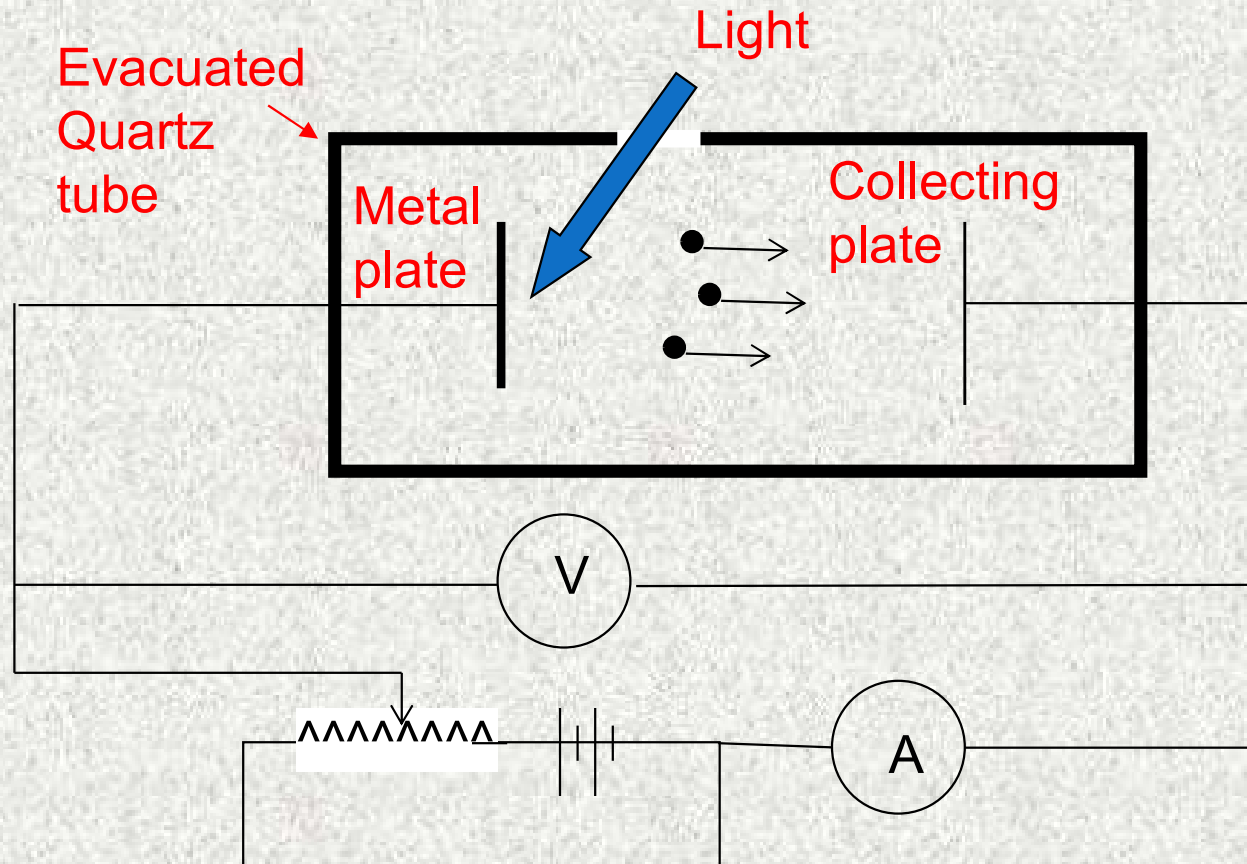
- Energy is not continuous, but comes in small but discrete units (Quanta) $h\nu$
- The elementary particles behave both like particles and like waves.
- The movement of these particles is inherently random.
- It is physically impossible to know both the position and the momentum of a particle at the same time.
- The atomic world is nothing like the world we live in.

The Planck's constant (h) characterizes Quantum Physics.

$$h = 6.625 \times 10^{-34} \text{ Joule.sec}$$

Photo Electric Effect

The emission of electrons from a metal plate when illuminated by light of suitable wavelength is called photoelectric effect. The emitted electrons are called 'photo electrons' and current so produced is called photo-electric current.



This effect is based on the principle of conservation of energy.

(A) Terms related to photoelectric effect

(i) **Work function (or threshold energy) (W_0):** The minimum energy of incident radiation, required to eject the electrons from metallic surface is defined as work function of that surface.

$$W_0 = h\nu_0 = \frac{hc}{\lambda_0} \text{ Joules ; } \nu_0 = \text{Threshold frequency; } \lambda_0 = \text{Threshold wavelength}$$

$$\text{Work function in electron volt } W_0 (eV) = \frac{hc}{e\lambda_0} = \frac{12375}{\lambda_0 (\text{\AA})}$$

Note : ☐ By coating the metal surface with a layer of barium oxide it's work function is lowered.

(ii) **Threshold frequency (ν_0):** The minimum frequency of incident radiations required to eject the electron from metal surface is defined as threshold frequency.

If incident frequency $\nu < \nu_0 \Rightarrow$ No photoelectron emission

(iii) **Threshold wavelength (λ_0):** The maximum wavelength of incident radiations required to eject the electrons from a metallic surface is defined as threshold wavelength.

If incident wavelength $\lambda > \lambda_0 \Rightarrow$ No photoelectron emission

(B) Einstein's photoelectric equation

According to Einstein, photoelectric effect is the result of one to one inelastic collision between photon and electron in which photon is completely absorbed. So if an electron in a metal absorbs a photon of energy $E (= h\nu)$, it uses the energy in three following ways.

(i) Some energy (say W) is used in shifting the electron from interior to the surface of the metal.

(ii) Some energy (say W_0) is used in making the surface electron free from the metal.

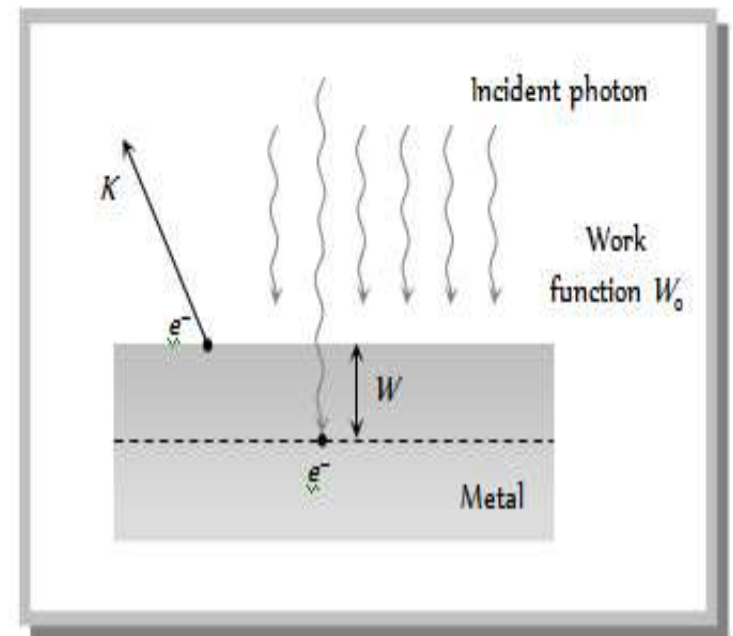
(iii) Rest energy will appear as kinetic energy (K) of the emitted photoelectrons.

$$\text{Hence } E = W + W_0 + K$$

For the electrons emitting from surface $W = 0$ so kinetic energy of emitted electron will be max.

$$\text{Hence } E = W_0 + K_{\max};$$

This is the Einstein's photoelectric equation



(3) Experimental arrangement to observe photoelectric effect

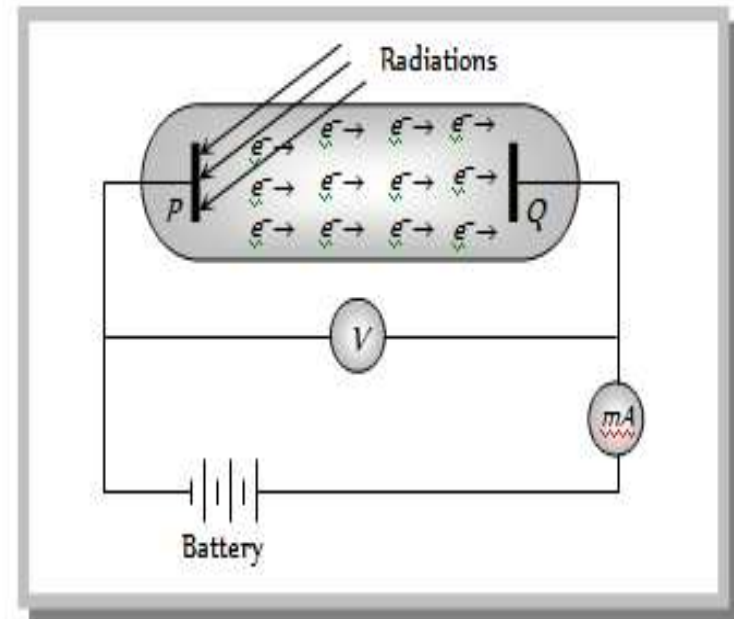
When light radiations of suitable frequency (or suitable wavelength and suitable energy) fall on plate P , photoelectrons are emitted from P .

(i) If plate Q is at zero potential *w.r.t.* P , very small current flows in the circuit because of some electrons of high kinetic energy are reaching to plate Q , but this current has no practical utility.

(ii) If plate Q is kept at positive potential *w.r.t.* P current starts flowing through the circuit because more electrons are able to reach upto plate Q .

(iii) As the positive potential of plate Q increases, current through the circuit increases but after some time constant current flows through the circuit even positive potential of plate Q is still increasing, because at this condition all the electrons emitted from plate P are already reached up to plate Q . This constant current is called **saturation current**.

(iv) To increase the photoelectric current further we will have to increase the intensity of incident light.



Photoelectric current (i) depends upon

- (a) Potential difference between electrodes (till saturation)
- (b) Intensity of incident light (I)
- (c) Nature of surface of metal



(v) To decrease the photoelectric current plate Q is maintained at negative potential *w.r.t.* P , as the anode Q is made more and more negative, fewer and fewer electrons will reach the cathode and the photoelectric current decreases.

(vi) At a particular negative potential of plate Q no electron will reach the plate Q and the current will become zero, this negative potential is called **stopping potential** denoted by V_0 .

(vii) If we increase further the energy of incident light, kinetic energy of photoelectrons increases and more negative potential should be applied to stop the electrons to reach up to plate Q . hence $eV_0 = K_{max}$.

Note: □ Stopping potential depends only upon frequency or wavelength or energy of incident radiation. It doesn't depend upon intensity of light.

We must remember that intensity of incident light radiation is inversely proportional to the square of distance between source of light and photosensitive plate P i.e., $I \propto \frac{1}{d^2}$ so $I \propto i \propto \frac{1}{d^2}$

Important formulae

$$\Rightarrow h\nu = h\nu_0 + K_{\max}$$

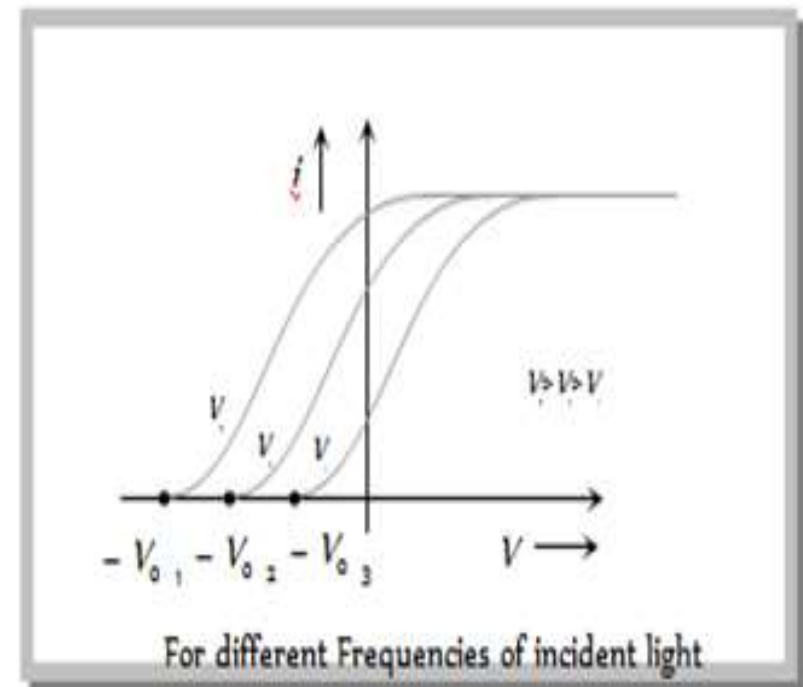
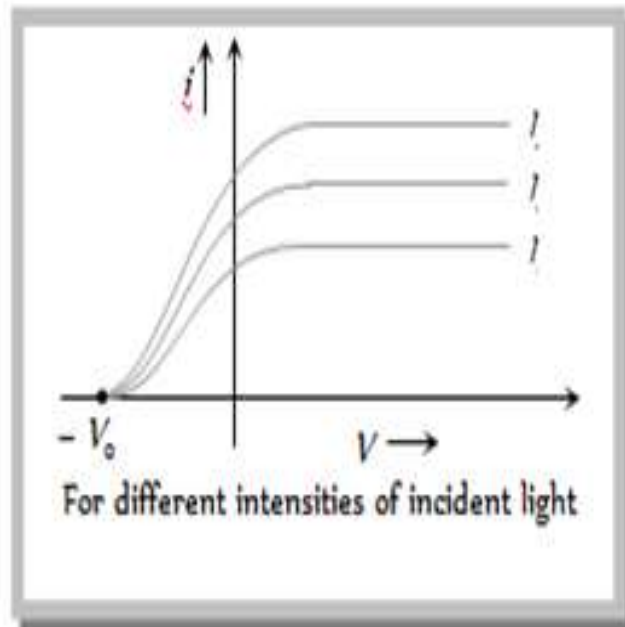
$$\Rightarrow K_{\max} = eV_0 = h(\nu - \nu_0) \Rightarrow \frac{1}{2}mv_{\max}^2 = h(\nu - \nu_0) \Rightarrow v_{\max} = \sqrt{\frac{2h(\nu - \nu_0)}{m}}$$

$$\Rightarrow K_{\max} = \frac{1}{2}mv_{\max}^2 = eV_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) = hc\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right) \Rightarrow v_{\max} = \sqrt{\frac{2hc}{m} \frac{(\lambda - \lambda_0)}{\lambda\lambda_0}}$$

$$\Rightarrow V_0 = \frac{h}{e}(\nu - \nu_0) = \frac{hc}{e}\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) = 12375 \left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

(4) Different graphs

Graph between potential difference between the plates P and Q and photoelectric current

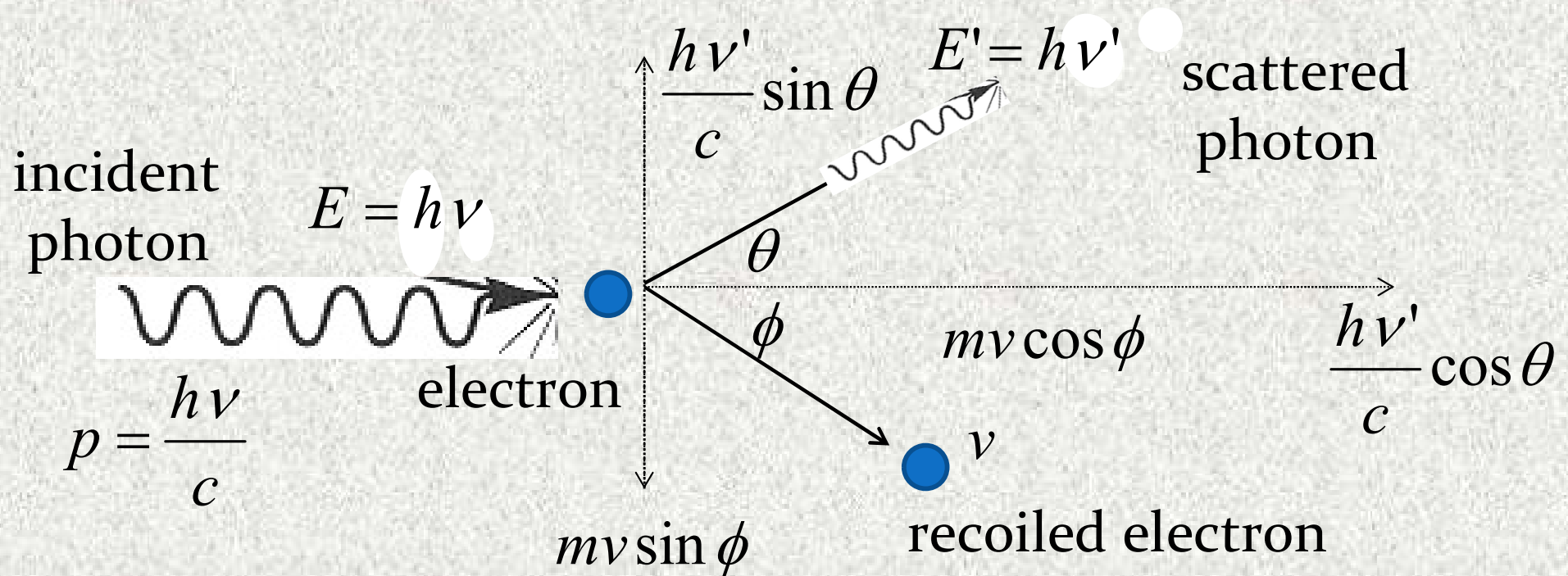


Compton Effect; Prof. A. H. Compton 1921-23

When a monochromatic beam of X-rays is incident on a material, then in addition to the scattered electron, two type of radiations are observed-

- 1- Original incident radiation (unmodified radiation)
- 2- A radiation of higher wavelength or lower energy (modified radiation)

This phenomenon is known as Compton Scattering.



From Theory of Relativity, total energy of the recoiled electron with $v \sim c$ is

$$E = mc^2 = K + m_o c^2$$

$$K = mc^2 - m_o c^2$$

$$K = \frac{m_o c^2}{\sqrt{1 - v^2/c^2}} - m_o c^2$$

$$K = m_o c^2 \left[\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right]$$

Similarly, momentum of recoiled electron is

$$mv = \frac{m_o v}{\sqrt{1 - v^2/c^2}}$$

Now from Energy Conversation

$$h\nu = h\nu' + m_0 c^2 \left[\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right] \quad (\text{i})$$

From Momentum Conversation

$$\frac{h\nu}{c} = \frac{h\nu'}{c} \cos \theta + \frac{m_0 v}{\sqrt{1 - v^2/c^2}} \cos \phi \quad (\text{ii}) \quad \text{along x-axis}$$

and

$$0 = \frac{h\nu'}{c} \sin \theta - \frac{m_0 v}{\sqrt{1 - v^2/c^2}} \sin \phi \quad (\text{iii}) \quad \text{along y-axis}$$

Rearranging (ii) and squaring both sides

$$\left(\frac{h\nu}{c} - \frac{h\nu'}{c} \cos \theta \right)^2 = \frac{m_o^2 v^2}{1 - v^2/c^2} \cos^2 \phi \quad (\text{iv})$$

Rearranging (iii) and squaring both sides

$$\left(\frac{h\nu'}{c} \sin \theta \right)^2 = \frac{m_o^2 v^2}{1 - v^2/c^2} \sin^2 \phi \quad (\text{v})$$

Adding (iv) and (v)

$$\left(\frac{h\nu}{c} \right)^2 + \left(\frac{h\nu'}{c} \right)^2 - \frac{2h^2\nu\nu'}{c^2} \cos \theta = \frac{m_o^2 v^2}{1 - v^2/c^2} \quad (\text{vi})$$

From equation (i) (Energy Conservation equation)

$$\frac{h\nu}{c} - \frac{h\nu'}{c} + m_o c = \frac{m_o c}{\sqrt{1 - v^2/c^2}}$$

On squaring, we get

$$\left(\frac{h\nu}{c}\right)^2 + \left(\frac{h\nu'}{c}\right)^2 + m_o^2 c^2 - \frac{2h^2 \nu \nu'}{c^2} + 2hm_o(\nu - \nu') = \frac{m_o^2 c^2}{1 - \nu^2/c^2} \quad (\text{vii})$$

Subtracting (vi) from (vii)

$$-\frac{2h^2 \nu \nu'}{c^2} (1 - \cos \theta) + 2hm_o(\nu - \nu') = 0$$

$$2hm_o(\nu - \nu') = \frac{2h^2 \nu \nu'}{c^2} (1 - \cos \theta)$$

$$m_o(\nu - \nu') = \frac{h \nu \nu'}{c^2} (1 - \cos \theta)$$

But $\nu = \frac{c}{\lambda}$ and $\nu' = \frac{c}{\lambda'}$ So,

$$m_o c \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = \frac{h}{\lambda \lambda'} (1 - \cos \theta)$$

$$m_o c \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right) = \frac{h}{\lambda \lambda'} (1 - \cos \theta)$$

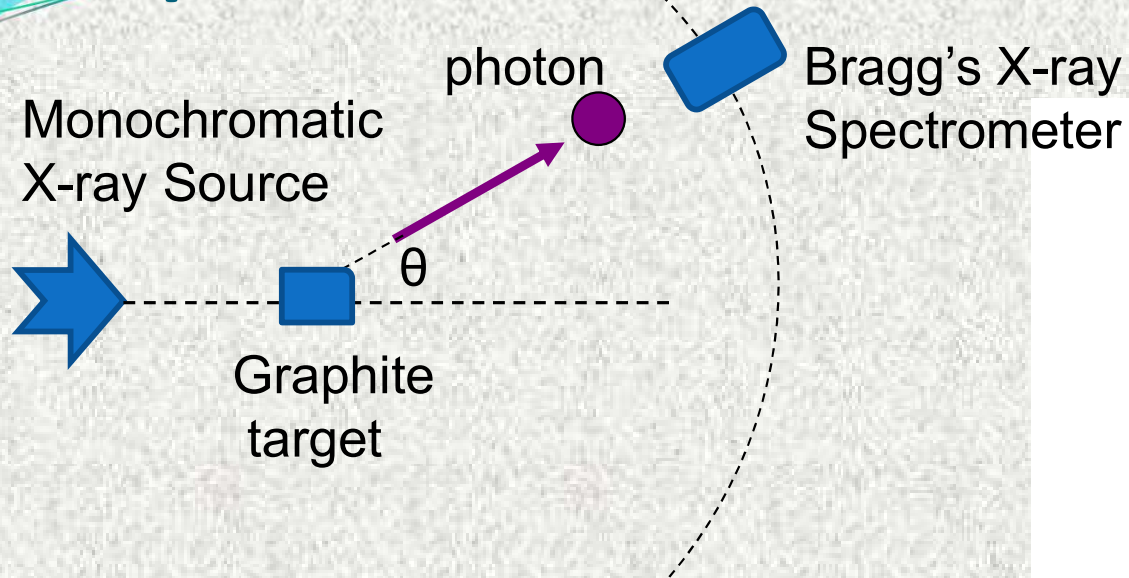
$$\lambda' - \lambda = \Delta \lambda = \frac{h}{m_o c} (1 - \cos \theta)$$

$\Delta \lambda$ is the Compton Shift.

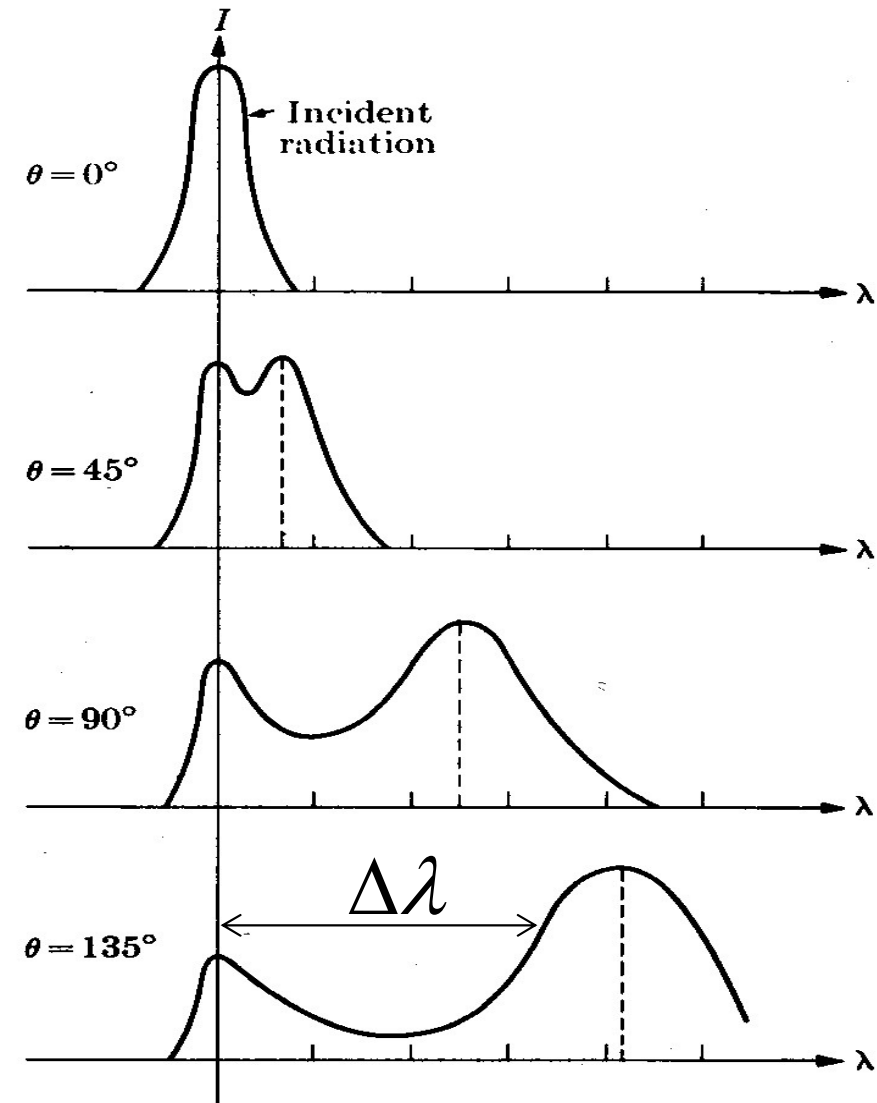
It neither depends on the incident wavelength nor on the scattering material. It only on the scattering angle i.e. θ

$\frac{h}{m_o c}$ is called the Compton wavelength of the electron and its value is 0.0243 Å.

Experimental Verification



1. One peak is found at same position. This is unmodified radiation
2. Other peak is found at higher wavelength. This is modified signal of low energy.
3. $\Delta\lambda$ increases with increase in θ .



Compton effect can't observed in Visible Light

$$\Delta\lambda = \frac{h}{m_o c} (1 - \cos \theta) = 0.0243 (1 - \cos \theta) \text{ \AA}$$

is maximum when $(1 - \cos \theta)$ is maximum i.e. 2.

$$0.05 \text{ \AA}$$

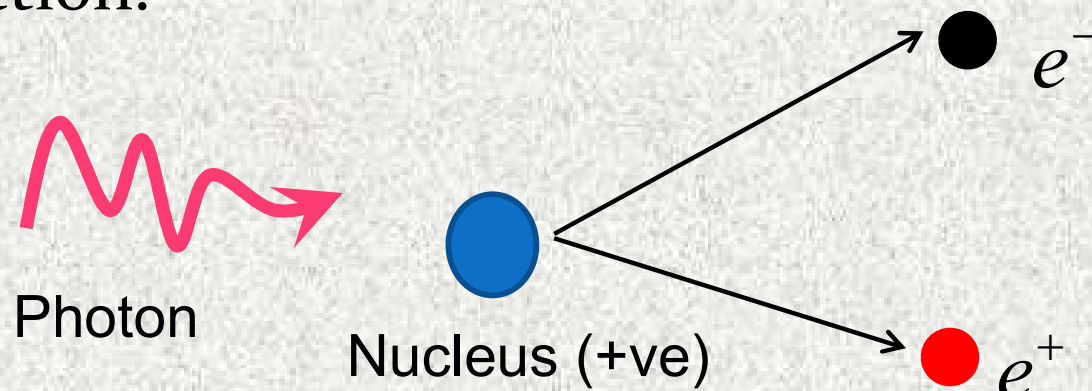
So Compton effect can be observed only for radiation having wavelength of few Å.

$$\text{For } \lambda = 1 \text{ \AA} \quad \Delta\lambda \sim 1\%$$

$$\text{For } \lambda = 5000 \text{ \AA} \quad \text{ } \sim 0.001\% \text{ (undetectable)}$$

Pair Production

When a photon (electromagnetic energy) of sufficient energy passes near the field of nucleus, it materializes into an electron and positron. This phenomenon is known as pair production.



In this process charge, energy and momentum remains conserved prior and after the production of pair.

The rest mass energy of an electron or positron is 0.511 MeV (according to $E = mc^2$).

The minimum energy required for pair production is 1.02 MeV.

Any additional photon energy becomes the kinetic energy of the electron and positron.

The corresponding maximum photon wavelength is 1.2 pm. Electromagnetic waves with such wavelengths are called gamma rays (γ).

Pair Annihilation

When an electron and positron interact with each other due to their opposite charge, both the particle can annihilate converting their mass into electromagnetic energy in the form of two γ - rays photon.

$$e^{-} + e^{+} \rightarrow \gamma + \gamma$$

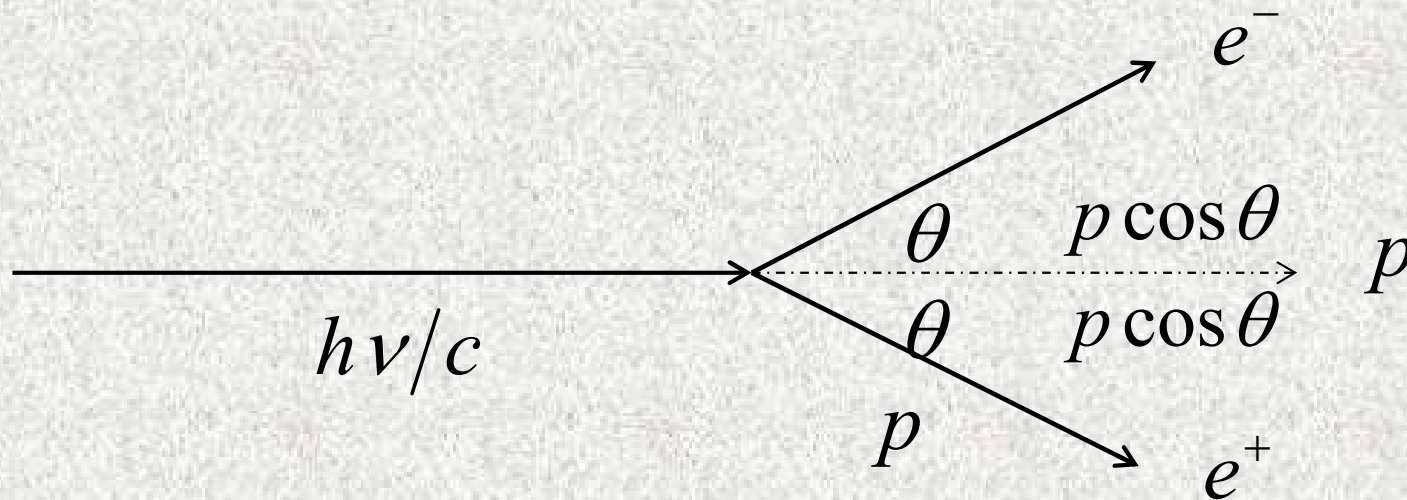
Charge, energy and momentum are again conversed. Two γ - photons are produced (each of energy 0.51 MeV plus half the K.E. of the particles) to conserve the momentum.

Pair production cannot occur in empty space

From conservation of energy

$$h\nu = 2m_0c^2\gamma$$

here m_0 is the rest mass and $\gamma = 1/\sqrt{1 - v^2/c^2}$



In the direction of motion of the photon, the momentum is conserved if

$$\frac{h\nu}{c} = 2p \cos \theta$$

$$h\nu = 2cp \cos \theta \quad (i)$$

Momentum of electron and positron is

$$p = m_o v \gamma$$

Equation (i) now becomes

$$h\nu = 2m_o c v \gamma \cos \theta$$

$$h\nu = 2m_o c^2 \gamma \left(\frac{v}{c} \right) \cos \theta$$

But $\frac{v}{c} < 1$ and $\cos \theta \leq 1$

$$h\nu < 2m_o c^2 \gamma$$

But conservation of energy requires that

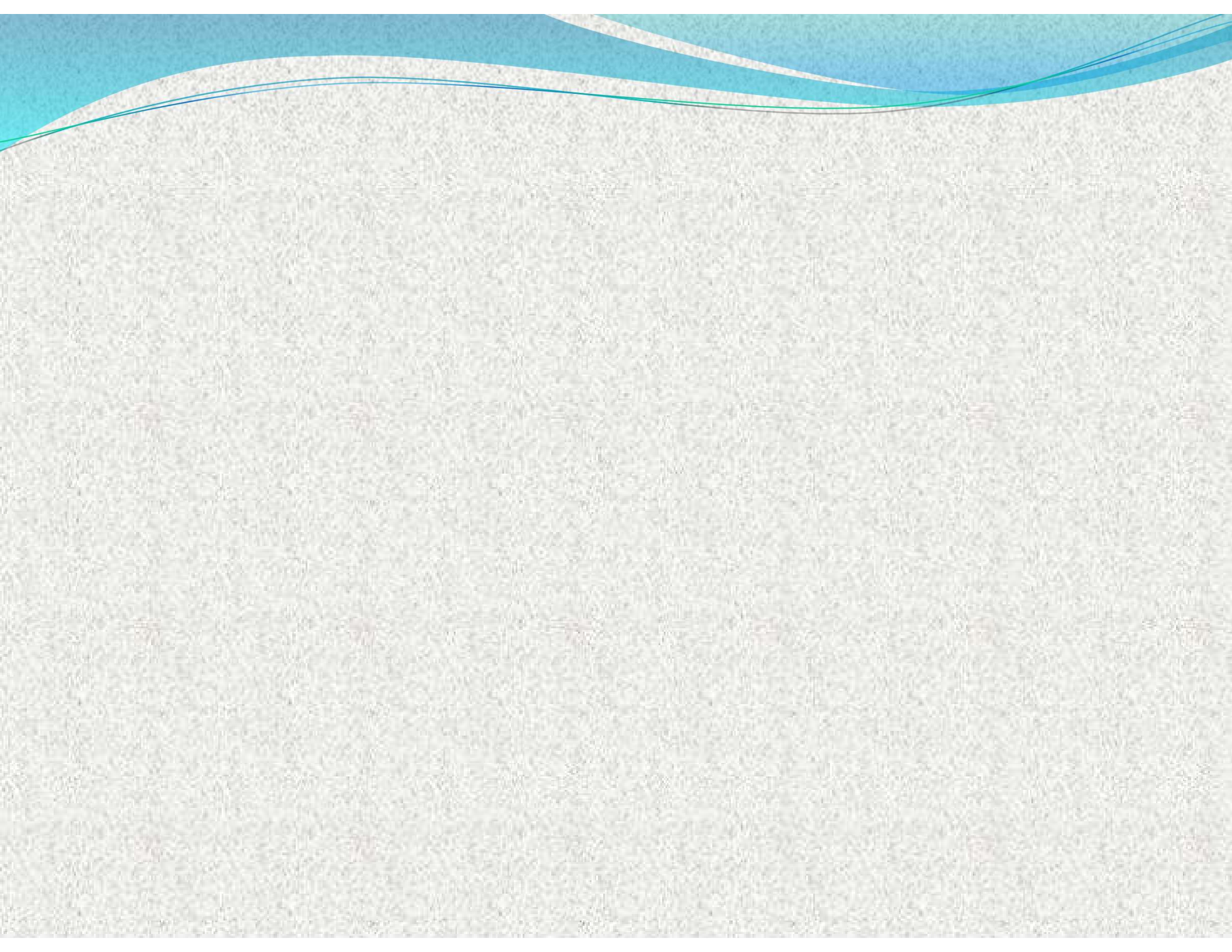
$$h\nu = 2m_0c^2\gamma$$

Hence it is impossible for pair production to conserve both the energy and momentum unless some other object is involved in the process to carry away part of the initial photon momentum. Therefore pair production cannot occur in empty space.

Photons

Einstein postulated the existence of a particle called a photon, to explain detailed results of photoelectric experiment. Photon has zero rest mass, travels at speed of light.

$$E_p = h\nu = \frac{hc}{\lambda}$$



Wave Particle Duality

Light can exhibit both kind of nature of waves and particles so the light shows wave-particle dual nature.

In some cases like interference, diffraction and polarization it behaves as wave while in other cases like photoelectric and Compton effect it behaves as particles (photon).

De Broglie Waves

Not only the light but every material particle such as electron, proton or even the heavier object exhibits wave-particle dual nature.

De-Broglie proposed that a moving particle, irrespective of its nature, has waves associated with it. These waves are called “**matter waves**” or “**De-Broglie waves**”.

Waves of Probability ????

Derivation of Formula

Since, Energy of a photon is

$$E = h \nu$$

For a particle, say photon of mass, m

$$E = mc^2$$

$$mc^2 = h\nu$$

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

Suppose a particle of mass, m is moving with velocity, v then the wavelength associated with it can be given by

$$\lambda = \frac{h}{mv} \quad \text{or} \quad \lambda = \frac{h}{p}$$

- (i) If $v = 0 \Rightarrow \lambda = \infty$ means that waves are associated with **moving** material particles only.
- (ii) De-Broglie wave does not depend on whether the moving particle is charged or uncharged. It means matter waves are not electromagnetic in nature.

Wave Velocity or Phase Velocity

When a monochromatic wave travels through a medium, its velocity of advancement in the medium is called the wave velocity or phase velocity (V_p).

$$V_p = \frac{\omega}{k}$$

where $\omega = 2\pi\nu$ is the angular frequency

and $k = \frac{2\pi}{\lambda}$ is the wave number.

Group Velocity

Generally, we come across with the pulses rather than monochromatic waves. A pulse consists of a number of waves differing slightly from one another in frequency.

So the group velocity is the velocity with which the energy in the group is transmitted (V_g).

The individual waves travel “inside” the group with their phase velocities.

$$V_g = \frac{d\omega}{dk}$$

Relation between Phase and Group Velocity

$$V_g = \frac{d\omega}{dk} = \frac{d}{dk}(kV_p)$$

$$V_g = V_p + k \frac{dV_p}{dk}$$

$$V_g = V_p + \frac{2\pi}{\lambda} \frac{dV_p}{d(2\pi/\lambda)}$$

$$V_g = V_p + \frac{1}{\lambda} \frac{dV_p}{d(1/\lambda)}$$

$$V_g = V_p + \frac{1}{\lambda} \frac{dV_p}{\left(-\frac{1}{\lambda^2} d\lambda\right)}$$

$$V_g = V_p - \lambda \frac{dV_p}{d\lambda}$$

In a non-dispersive medium - such as empty space

(in a non-dispersive medium if you change the wave-frequency, the wave-length adjusts such that v remains constant)

$$\frac{\omega}{k} = \text{constant} = V_p$$

$$\Rightarrow \frac{dV_p}{d\lambda} = 0$$

$$\Rightarrow V_g = V_p$$

In a Dispersive medium V_p depends on frequency

$$\text{i.e. } \frac{\omega}{k} \neq \text{constant}$$

So, $\lambda \frac{dV_p}{d\lambda}$ is positive generally (not always).

$$\Rightarrow V_g < V_p \quad \text{generally}$$

$$V_g = V_p - \lambda \frac{dV_p}{d\lambda}$$

Phase Velocity of De-Broglie's waves

According to De-Broglie's hypothesis of matter waves

$$\lambda = \frac{h}{mv}$$

wave number $k = \frac{2\pi}{\lambda} = \frac{2\pi mv}{h}$ (i)

If a particle has energy E , then corresponding wave will have frequency

$$\nu = \frac{E}{h}$$

then angular frequency will be $\omega = 2\pi\nu = \frac{2\pi E}{h}$

$$\omega = \frac{2\pi mc^2}{h} \quad (\text{ii})$$

Dividing (ii) by (i)

$$\frac{\omega}{k} = \frac{2\pi mc^2}{h} \times \frac{h}{2\pi mv}$$

$$V_p = \frac{c^2}{v}$$

But v is always $< c$ (velocity of light)

(i) Velocity of De-Broglie's waves $V_p > c$ (not acceptable)

(ii) De-Broglie's waves (V_p) will move faster than the particle velocity (v) and hence the waves would leave the particle behind.

Group Velocity of De-Broglie's waves

The discrepancy is resolved by postulating that a **moving particle is associated with a “wave packet” or “wave group”**, rather than a single wave-train.

A wave group having wavelength λ is composed of a number of component waves with slightly different wavelengths in the neighborhood of λ .

Suppose a particle of rest mass m_0 moving with velocity v then associated matter wave will have

$$\omega = \frac{2\pi mc^2}{h} \quad \text{and} \quad k = \frac{2\pi mv}{h} \quad \text{where} \quad m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\omega = \frac{2\pi m_o c^2}{h\sqrt{1-v^2/c^2}} \quad \text{and} \quad k = \frac{2\pi m_o v}{h\sqrt{1-v^2/c^2}}$$

On differentiating w.r.t. velocity, v

$$\frac{d\omega}{dv} = \frac{2\pi m_o v}{h(1-v^2/c^2)^{3/2}} \quad (\text{i})$$

$$\frac{dk}{dv} = \frac{2\pi m_o}{h(1-v^2/c^2)^{3/2}} \quad (\text{ii})$$

Dividing (i) by (ii)

$$\frac{d\omega}{dv} \cdot \frac{dv}{dk} = \frac{2\pi m_0 v}{2\pi m_0}$$

$$\frac{d\omega}{dk} = v = V_g$$

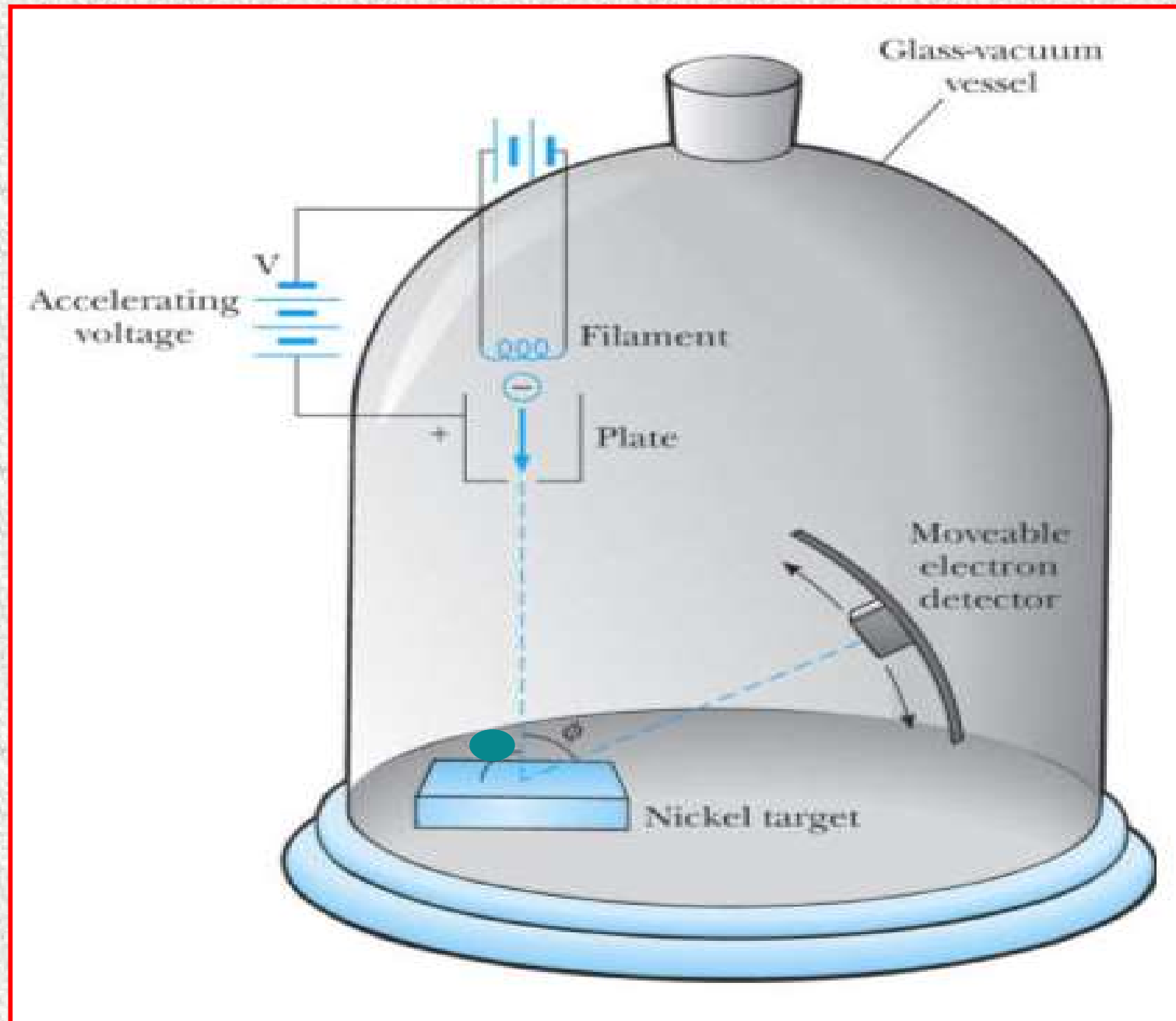
Wave group associated with a moving particle also moves with the velocity of the particle.

Moving particle $\equiv$ wave packet or wave group

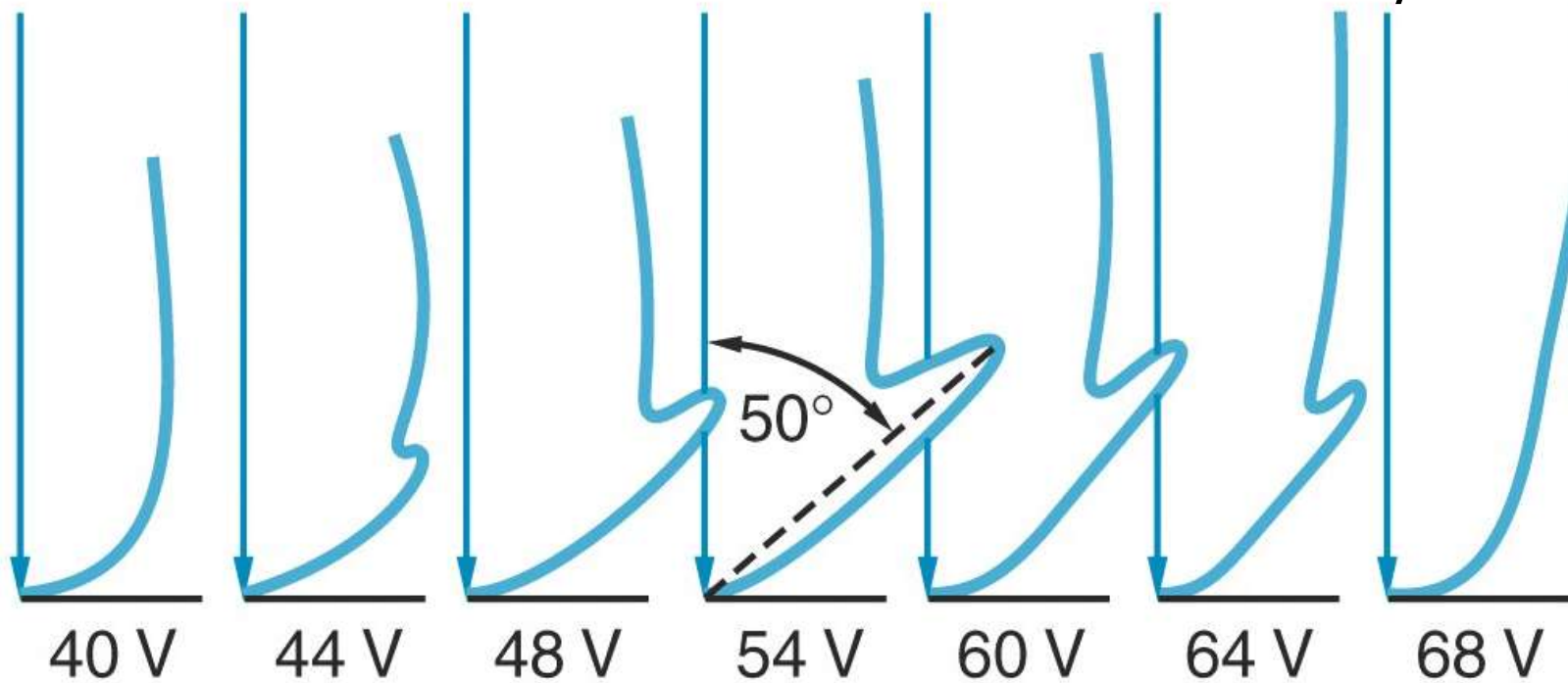
Davisson & Germer experiment of electron diffraction

- If particles have a wave nature, then under appropriate conditions, they should exhibit diffraction
- Davisson & Germer measured the wavelength of electrons
- This provided experimental confirmation of the matter waves proposed by de Broglie

Davisson and Germer Experiment



Incident Beam



Current vs accelerating voltage has a maximum (a bump or kink noticed in the graph), i.e. the highest number of electrons is scattered in a specific direction.

The bump becomes most prominent for 54 V at $\phi \sim 50^\circ$

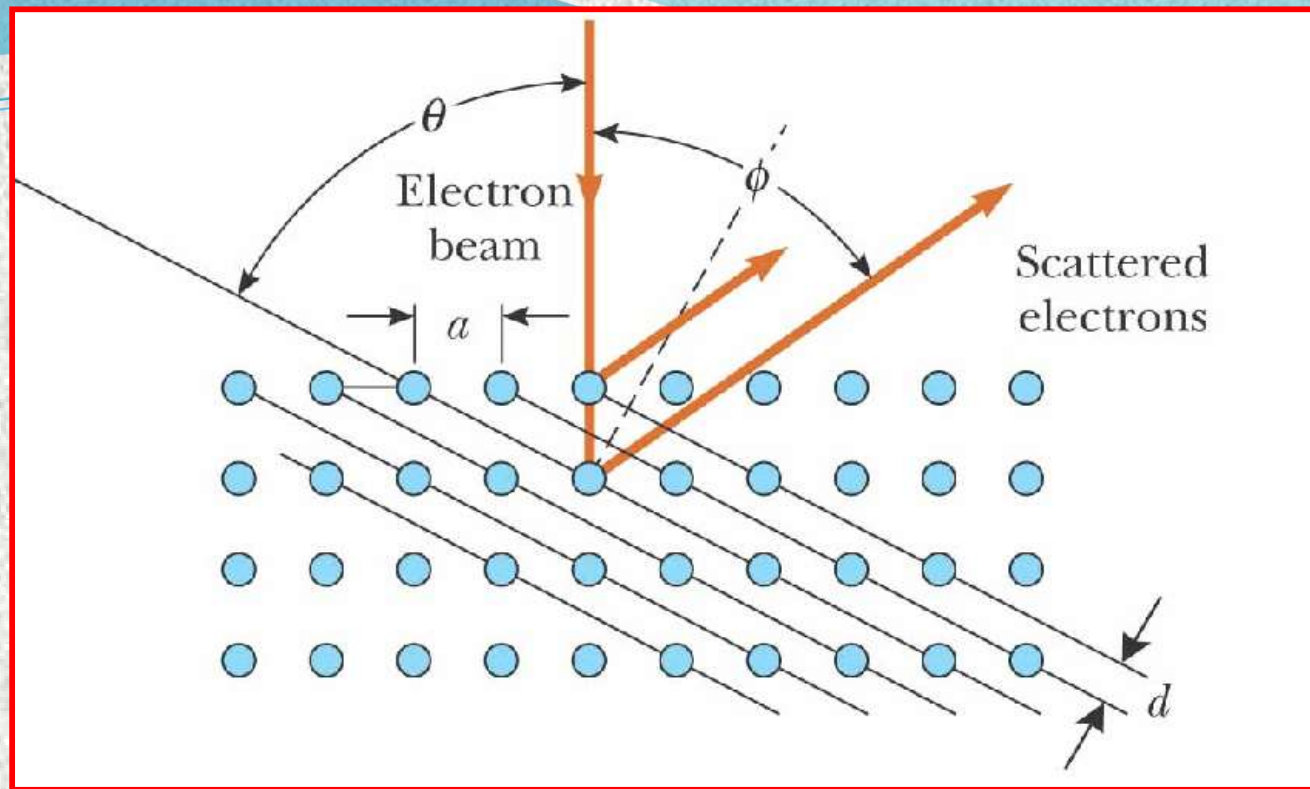
According to de Broglie, the wavelength associated with an electron accelerated through V volts is

$$\lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

Hence the wavelength for 54 V electron

$$\lambda = \frac{12.28}{\sqrt{54}} = 1.67 \text{ \AA}$$

From X-ray analysis we know that the nickel crystal acts as a plane diffraction grating with grating space $d = 0.91 \text{ \AA}$



Here the diffraction angle, $\phi \sim 50^\circ$

The angle of incidence relative to the family of Bragg's plane

$$\theta = \left(\frac{180^\circ - 50^\circ}{2} \right) = 65^\circ$$

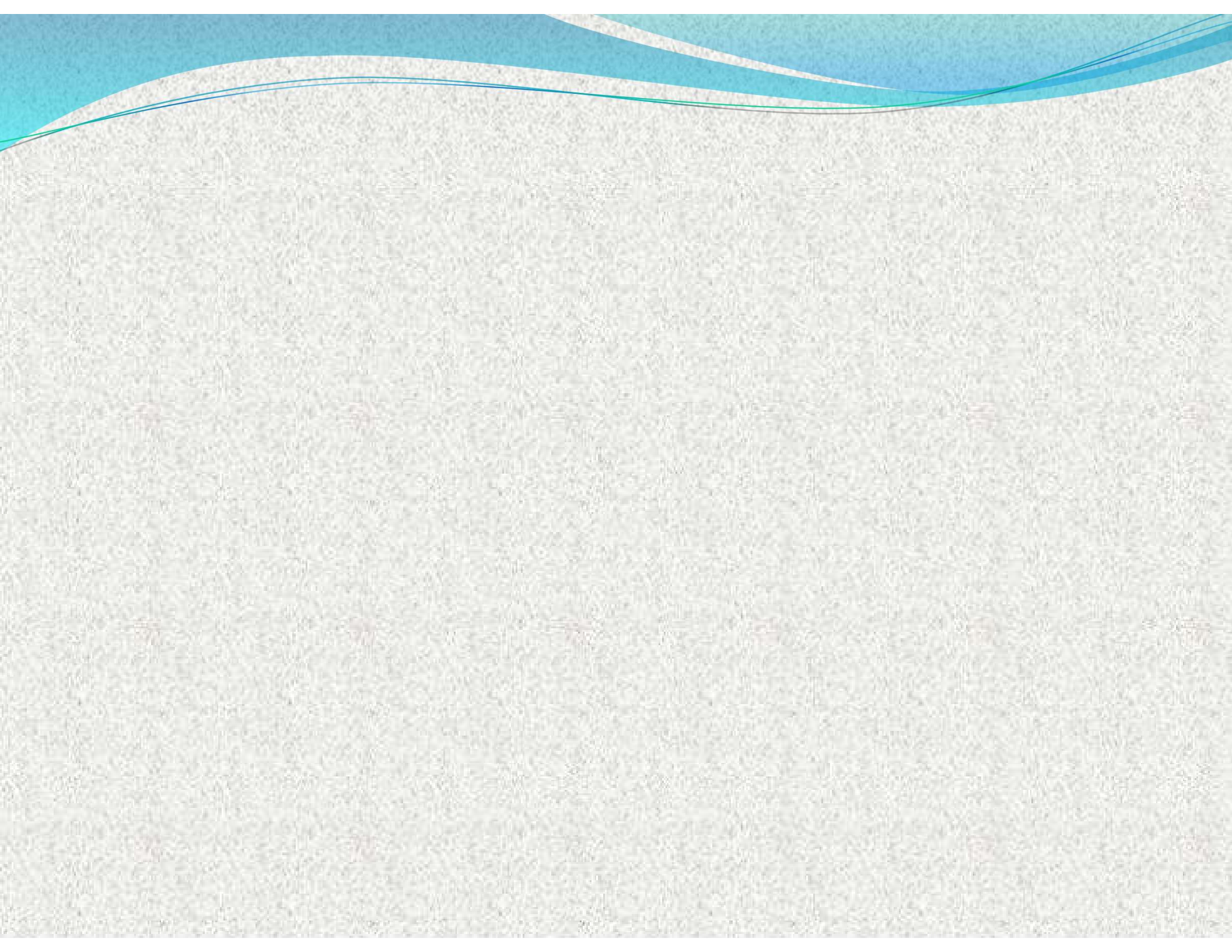
From the Bragg's equation

$$\lambda = 2d \sin \theta$$

$$\lambda = 2 \times (0.91 \text{ \AA}) \times \sin 65^\circ = 1.65 \text{ \AA}$$

which is equivalent to the λ calculated by de-Broglie's hypothesis.

It confirms the wavelike nature of electrons



Wave function

Wave function, ψ is a quantity associated with a moving particle. It is a complex quantity.

$|\Psi|^2$ is proportional to the probability of finding a particle at a particular point at a particular time. It is the probability density.

$$|\psi|^2 = \psi^* \psi$$

ψ is the probability amplitude.

Thus if $\psi = A + iB$ then $\psi^* = A - iB$

$$\Rightarrow |\psi|^2 = \psi^* \psi = A^2 - i^2 B^2 = A^2 + B^2$$

Normalization

$|\Psi|^2$ is the probability density.

The probability of finding the particle within an element of volume dV

$$|\psi|^2 dV$$

Since the particle is definitely be somewhere, so

$$\int_{-\infty}^{\infty} |\psi|^2 dV = 1$$

A wave function always follows the normalization equation.

Properties of wave function

1. It must be finite everywhere.

If ψ is infinite for a particular point, it mean an infinite large probability of finding the particles at that point. This would violates the uncertainty principle.

2. It must be single valued.

If ψ has more than one value at any point, it mean more than one value of probability of finding the particle at that point which is obviously ridiculous.

3. It must be continuous and have a continuous first derivative everywhere.

$$\frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial z} \quad \text{must be continuous}$$

4. It must be normalizable.

Linearity and Superposition

If ψ_1 and ψ_2 are two solutions of any Schrodinger equation of a system, then linear combination of ψ_1 and ψ_2 will also be a solution of the equation..

$$\psi = a_1\psi_1 + a_2\psi_2 \quad \text{is also a solution}$$

Here a_1 & a_2 are constants

Above equation suggests:

- (i) The linear property of Schrodinger equation
- (ii) ψ_1 and ψ_2 follow the superposition principle

If P_1 is the probability density corresponding to ψ_1 and P_2 is the probability density corresponding to ψ_2

Then $\psi \rightarrow \psi_1 + \psi_2$ due to superposition principle

Total probability will be

$$\begin{aligned}
 P &= |\psi|^2 = |\psi_1 + \psi_2|^2 \\
 &= (\psi_1 + \psi_2)^* (\psi_1 + \psi_2) = (\psi_1^* + \psi_2^*)(\psi_1 + \psi_2) \\
 &= \psi_1^* \psi_1 + \psi_2^* \psi_2 + \psi_1^* \psi_2 + \psi_2^* \psi_1
 \end{aligned}$$

$$P = P_1 + P_2 + \psi_1^* \psi_2 + \psi_2^* \psi_1 \quad P \neq P_1 + P_2$$

Probability density can't be added linearly

Expectation values

Expectation value of any quantity $f(x)$ (which is a function of 'x'), is given by

$$\langle f(x) \rangle = \int_{-\infty}^{\infty} f(x) |\psi|^2 dx$$

Thus expectation value for position 'x' is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\psi|^2 dx$$

Expectation value is the value of 'x' we would obtain if we measured the positions of a large number of particles described by the same function at some instant 't' and then averaged the results.

Q. Find the expectation value of position of a particle having wave function $\psi = ax$ between $x = 0$ & 1 , $\psi = 0$ elsewhere.

Solution

$$\langle x \rangle = \int_0^1 x |\psi|^2 dx = a^2 \int_0^1 x^3 dx$$

$$= a^2 \left[\frac{x^4}{4} \right]_0^1$$

$$\langle x \rangle = \frac{a^2}{4}$$

Operators

An operator is a **rule** by means of which, from a given function we can find another function.

For a free particle

$$\psi = Ae^{\frac{-i}{\hbar}(Et - px)}$$

Then $\frac{\partial \psi}{\partial x} = \frac{i}{\hbar} p \psi$ Here $\hat{p} = \frac{\hbar}{i} \frac{\partial}{\partial x}$ (i)

is called the momentum operator

Similarly $\frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} E \psi$ Here $\hat{E} = i\hbar \frac{\partial}{\partial t}$ (ii)

is called the Total Energy operator

Equation (i) and (ii) are general results and their validity is the same as that of the Schrodinger equation.

Schrodinger's time dependent wave equation

Wave equation for a free particle moving in +x direction is

$$\psi = Ae^{\frac{-i}{\hbar}(Et - px)} \quad (i)$$

where E is the total energy and p is the momentum of the particle

Differentiating (i) twice w.r.t. x

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi \quad \Rightarrow \quad p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} \quad (ii)$$

Differentiating (i) w.r.t. t

$$\frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar} \psi \quad \Rightarrow \quad E \psi = i\hbar \frac{\partial \psi}{\partial t} \quad (iii)$$

For non-relativistic case

$$E = \text{K.E.} + \text{Potential Energy}$$

$$E = \frac{p^2}{2m} + V_{x,t}$$

$$\Rightarrow E\psi = \frac{p^2}{2m}\psi + V\psi \quad (\text{iv})$$

Using (ii) and (iii) in (iv)

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

This is the time dependent Schrodinger's wave equation for a particle in one dimension.

Schrodinger's time independent wave equation

One dimensional wave equation for the waves associated with a moving particle is

$$\psi = A \sin \frac{2\pi x}{\lambda} \quad (i)$$

where

ψ is the wave amplitude for a given x .

A is the maximum amplitude.

λ is the wavelength

From (i)

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{\lambda^2} A \sin \frac{2\pi x}{\lambda}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{\lambda^2} \psi \quad (ii)$$

$$\lambda = \frac{h}{m_o v} \Rightarrow \frac{1}{\lambda^2} = \frac{m_o^2 v^2}{h^2} = \frac{2m_o \left(\frac{1}{2} m_o v^2 \right)}{h^2} \text{ so, } \frac{1}{\lambda^2} = \frac{2m_o K}{h^2}$$

where K is the K.E. for the non-relativistic case

Suppose E is the total energy of the particle

and V is the potential energy of the particle

$$\frac{1}{\lambda^2} = \frac{2m_o}{h^2} (E - V) \quad (\text{iii})$$

Equation (ii) now becomes

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{4\pi^2}{h^2} 2m_o (E - V) \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m_o}{\hbar^2} (E - V) \psi = 0$$

This is the time independent (steady state) Schrodinger's wave equation for a particle of mass m_o , total energy E , potential energy V , moving along the x-axis.

If the particle is moving in 3-dimensional space then

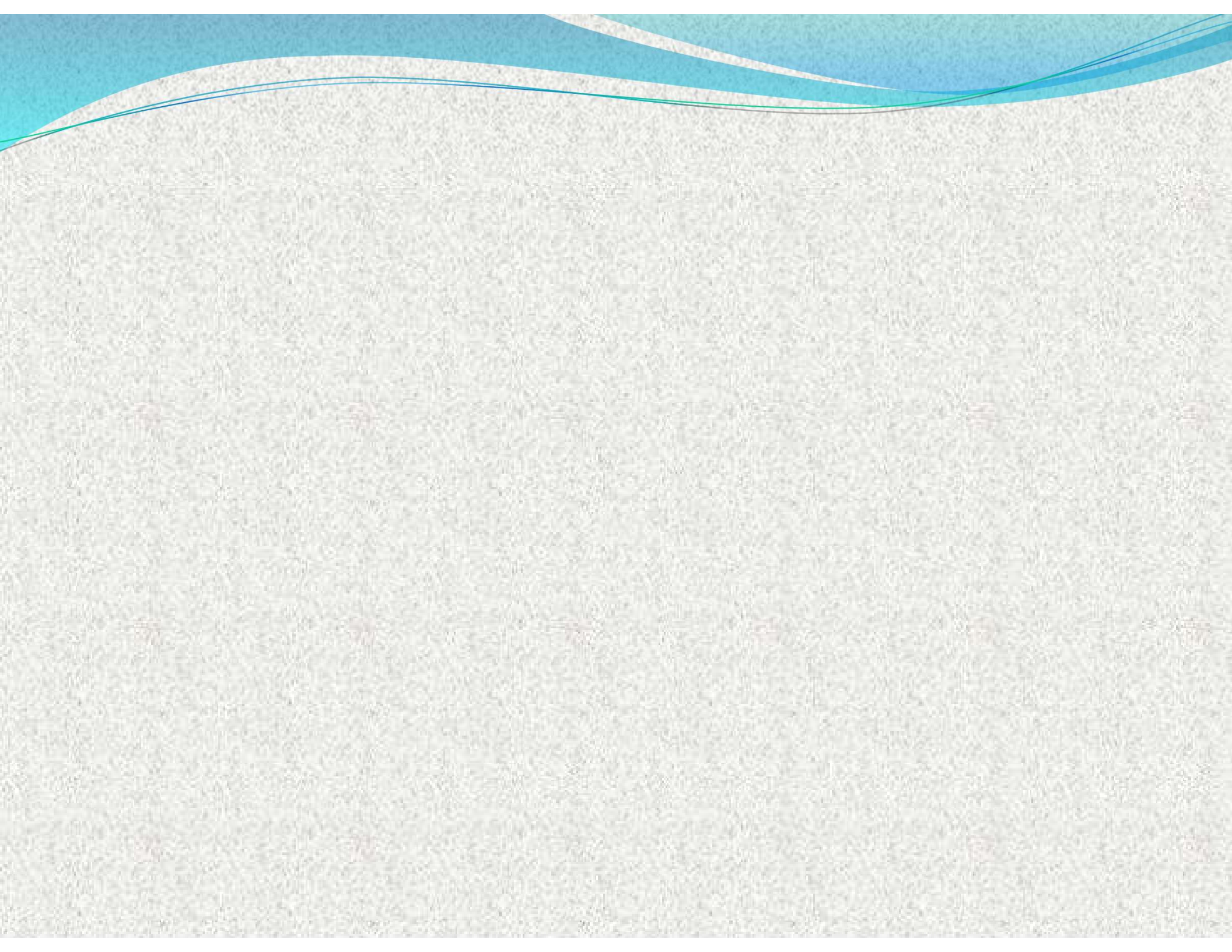
$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m_o}{\hbar^2} (E - V) \psi = 0$$

$$\nabla^2 \psi + \frac{2m_o}{\hbar^2} (E - V) \psi = 0$$

This is the time independent (steady state) Schrodinger's wave equation for a particle in 3-dimensional space.

For a free particle $V = 0$, so the Schrodinger equation for a free particle

$$\nabla^2 \psi + \frac{2m_o}{\hbar^2} E \psi = 0$$



Heisenberg Uncertainty Principle

It states that only **one** of the “**position**” or “**momentum**” can be measured accurately at a single moment within the instrumental limit.

$\Delta x \rightarrow$ uncertainty in position

$\Delta p_x \rightarrow$ uncertainty in momentum

$$\text{then } \Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \quad \therefore \hbar = \frac{h}{2\pi}$$

The product of Δx & Δp_x of an object is greater than or equal to $\frac{\hbar}{2}$

If Δx is measured accurately i.e. $\Delta x \rightarrow 0 \Rightarrow \Delta p_x \rightarrow \infty$

The principle applies to all **canonically conjugate pairs** of quantities in which measurement of one quantity affects the capacity to measure the other.

Like, energy E and time t. $\Delta E \Delta t \geq \frac{\hbar}{2}$

and angular momentum L and angular position θ $\Delta L \Delta \theta \geq \frac{\hbar}{2}$

Applications of Heisenberg Uncertainty Principle

Determination of the position of a particle by a microscope

Suppose we want to determine accurately the position and momentum of an electron along x-axis using an ideal microscope free from all mechanical and optical defects.

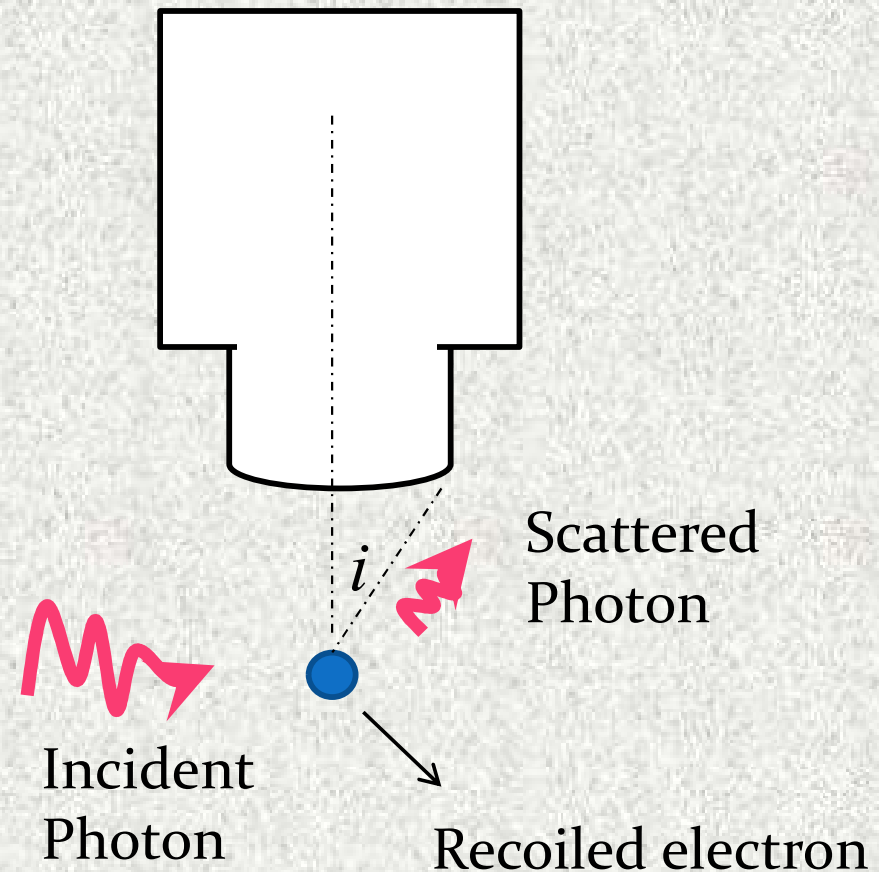
The limit of resolution of the microscope is

$$2d \sin \theta = n \lambda$$

$$\Delta x = \frac{\lambda}{2 \sin i}$$

here i is semi-vertex angle of the cone of rays entering the objective lens of the microscope.

Δx is the order of uncertainty in the x-component of the position of the electron.



We can't measure the momentum of the electron prior to illumination. So there is uncertainty in the measurement of momentum of the electron.

The scattered photon can enter the microscope anywhere between the angular range $+i$ to $-i$.

The momentum of the scattered photon is (de-Broglie) $p = \frac{h}{\lambda}$

Its x-component can be given as $\Delta p_x = \frac{2h}{\lambda} \sin i$

The x-component of the momentum of the recoiling electron has the same uncertainty, Δp_x (conservation of momentum)

The product of the uncertainties in the x-components of position and momentum for the electron is

$$\Delta x \cdot \Delta p_x = \frac{\lambda}{2 \sin i} \times \frac{2h}{\lambda} \sin i \qquad \Delta x \cdot \Delta p_x = h > \frac{\hbar}{2}$$

This is in agreement with the uncertainty relation.

Non-existence of electron in nucleus

Order of radius of an atom $\sim 5 \times 10^{-15} \text{ m}$

If electron exist in the nucleus then $(\Delta x)_{\text{max}} = 5 \times 10^{-15} \text{ m}$

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \quad (\Delta x)_{\text{max}} (\Delta p_x)_{\text{min}} = \frac{\hbar}{2}$$

$$(\Delta p_x)_{\text{min}} = \frac{\hbar}{2\Delta x} = 1.1 \times 10^{-20} \text{ kg.m.s}^{-1}$$

then
$$E = \frac{p^2}{2m} = 237.5 \text{ MeV}$$

Thus the kinetic energy of an electron must be greater than 237.5 MeV to be a part of nucleus

Experiments show that the electrons emitted by certain unstable nuclei don't have energy greater than 3-4 MeV.

Thus we can conclude that the electrons cannot be present within nuclei.

Energy of a particle in a box or infinite potential well

Let us consider a particle of mass m in infinite potential well of width L . The maximum uncertainty in the position of the particle may be

$$(\Delta x)_{\max} = L$$

From Uncertainty principle $\Delta x \Delta p_x \geq \frac{\hbar}{2}$

Here, we can write, minimum uncertainty in momentum-

$$\Delta p_x = \frac{\hbar}{(\Delta x)_{\max}} = \frac{\hbar}{L}$$

Since, the Kinetic Energy can be written as $KE = \frac{p^2}{2m}$

So, for present case

$$(KE)_{\min} = \frac{\hbar^2}{2mL^2}$$

This is the minimum kinetic energy of the particle in an infinite potential well

Particle in a Box: Eigenvalue

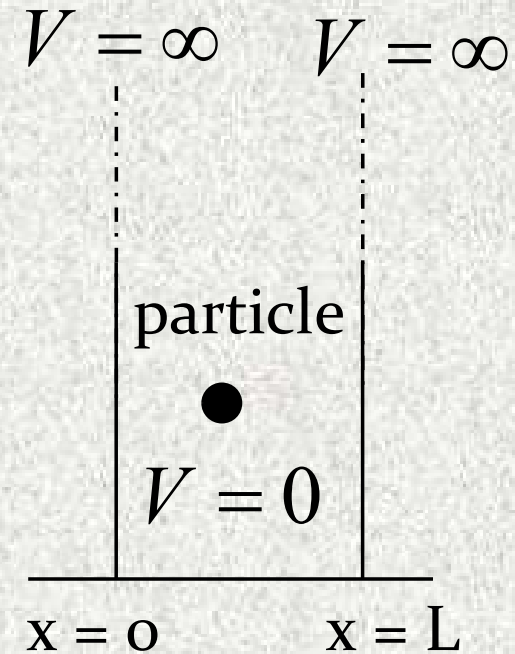
Consider a particle of rest mass m_o enclosed in a one-dimensional box (infinite potential well).

Boundary conditions for Potential V

$$V(x) = \begin{cases} 0 & \text{for } 0 < x < L \\ \infty & \text{for } 0 \geq x \geq L \end{cases}$$

Boundary conditions for wave function ψ

$$\psi = \begin{cases} 0 & \text{for } x = 0 \\ 0 & \text{for } x = L \end{cases}$$



Thus for a particle inside the box Schrodinger equation is

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m_o}{\hbar^2} E \psi = 0 \quad (i) \quad [\because V = 0 \text{ inside}]$$

$$\lambda = \frac{h}{p} = \frac{2\pi}{k} \quad (\text{k is the propagation constant})$$

$$\Rightarrow k = \frac{p}{\hbar} = \frac{\sqrt{2m_o E}}{\hbar}$$

$$\Rightarrow k^2 = \frac{2m_o E}{\hbar^2} \quad (\text{ii})$$

Equation (i) becomes

$$\frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0 \quad (\text{iii})$$

General solution of equation (iii) is

$$\psi(x) = A \sin kx + B \cos kx \quad (\text{iv})$$

Boundary condition says $\psi = 0$ when $x = 0$

$$\psi(0) = A \sin k.0 + B \cos k.0$$

$$0 = 0 + B.1 \quad \Rightarrow B = 0$$

Equation (iv) reduces to

$$\psi(x) = A \sin kx \quad (v)$$

Boundary condition says $\psi = 0$ when $x = L$

$$\psi(L) = A \sin k.L$$

$$0 = A \sin k.L$$

$$A \neq 0 \Rightarrow \sin k.L = 0$$

$$\Rightarrow \sin k.L = \sin n\pi$$

$$kL = n\pi$$

$$k = \frac{n\pi}{L} \quad (\text{vi})$$

Put this in Equation (v)

$$\psi(x) = A \sin \frac{n\pi x}{L}$$

(Where $n = 1, 2, 3, \dots$, BUT $n \neq 0$ as this gives $\psi = 0$ everywhere)

Put value of k from (vi) in (ii)

$$k^2 = \frac{2m_o E}{\hbar^2}$$

$$\left(\frac{n\pi}{L} \right)^2 = \frac{2m_o E}{\hbar^2}$$

$$\Rightarrow E = \frac{\hbar^2 k^2}{2m_o} = \frac{n^2 h^2}{8m_o L^2} \quad (\text{vii})$$

Where $n = 1, 2, 3, \dots$

Equation (vii) concludes

1. Energy of the particle inside the box can't be equal to zero.

The minimum energy of the particle is obtained for $n = 1$

$$E_1 = \frac{h^2}{8m_o L^2} \quad (\text{Zero Point Energy})$$

If $E_1 \rightarrow 0$ momentum $\rightarrow 0$ i.e. $\Delta p \rightarrow 0$

$$\Rightarrow \Delta x \rightarrow \infty$$

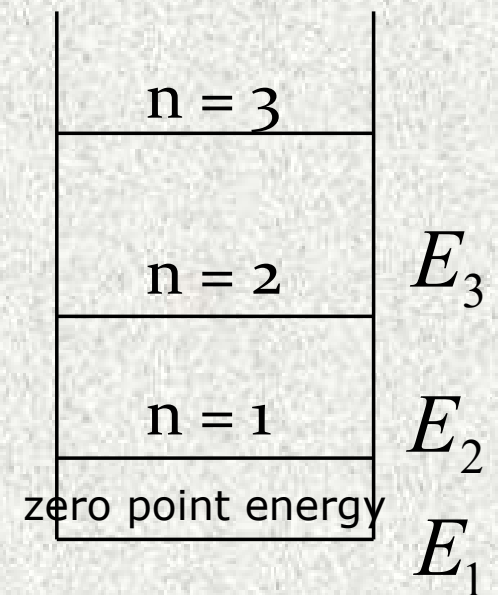
But $\Delta x_{\text{max}} = L$ since the particle is confined in the box of dimension L .

Thus zero value of zero point energy violates the Heisenberg's uncertainty principle and hence zero value is not acceptable.

2. All the energy values are not possible for a particle in potential well.

Energy is Quantized

3. E_n are the eigen values and 'n' is the quantum number.
4. Energy levels (E_n) are not equally spaced.



Particle in a Box: Eigenfunction

$$\psi_n(x) = A \sin \frac{n\pi x}{L}$$

Using Normalization condition

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1$$

$$A^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = 1$$

$$A^2 \left(\frac{L}{2} \right) = 1 \Rightarrow A = \sqrt{\frac{2}{L}}$$

The normalized eigen function of the particle are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi n x}{L}$$

Probability density figure suggest that:

1. There are some positions (nodes) in the box that will never be occupied by the particle.
2. For different energy levels the points of maximum probability are found at different positions in the box.

$|\psi_1|^2$ is maximum at $L/2$ (middle of the box)

$|\psi_2|^2$ is zero $L/2$.

Particle in a Three Dimensional Box

Eigen energy $E = E_x + E_y + E_z$

$$E = (n_x^2 + n_y^2 + n_z^2) \frac{h^2}{8mL^2}$$

Eigen function $\psi = \psi_x \psi_y \psi_z$

$$\psi = A_x A_y A_z \sin \frac{n_x \pi x}{L} \sin \frac{n_y \pi y}{L} \sin \frac{n_z \pi z}{L}$$

$$\psi = \left(\sqrt{\frac{2}{L}} \right)^3 \sin \frac{n_x \pi x}{L} \sin \frac{n_y \pi y}{L} \sin \frac{n_z \pi z}{L}$$

